

Q7) Using the standard free energies of formation (kJ mol^{-1}) :
 $\text{BaCO}_3(s)$, $\Delta G_f^\circ = -1139.0$ $\text{BaSO}_4(s)$, $\Delta G_f^\circ = -1353.0$ $\text{CO}_2(g)$, $\Delta G_f^\circ = -394.4$ $\text{SO}_3(g)$, $\Delta G_f^\circ = -370.0$ Calculate the standard free energy change, ΔG° , for the reaction: $\text{BaCO}_3(s) + \text{SO}_3(g) \rightarrow \text{BaSO}_4(s) + \text{CO}_2(g)$
 A. +189.6 kJ B. -238.4 kJ C. +472.4 kJ D. +1727.6 kJ
 E. -2516.4 kJ

Q8) The reaction, $\text{M}_2\text{O}_3(s) + \text{C}(s) \rightarrow \text{M}(s) + \text{CO}_2(g)$, is spontaneous at low temperatures but non-spontaneous at high temperatures. Assume that, the values of ΔH_f° and ΔS_f° are constant over this temperature range, we can deduce that:

- A. $\Delta H < 0$ and $\Delta S > 0$ B. $\Delta H < 0$ and $\Delta S < 0$ C. $\Delta H > 0$ and $\Delta S < 0$
 D. $\Delta H > 0$ and $\Delta S > 0$ E. $\Delta G = 0$ at 298 K

Q9) The Second Law of Thermodynamics states:

- A. $\Delta E_{\text{sys}} = q + w$
 B. The entropy of the universe increases for spontaneous processes.
 C. $\Delta S = 0$ for a pure element in its most stable state at standard conditions.
 D. $S = 0$ for a perfect crystal of a pure substance at 0 K.
 E. none of the above.

Q10) Which one of the following reactions has $\Delta S > 0$?

- A. $2 \text{H}(g) \rightarrow \text{H}_2(g)$
 B. $\text{NiCl}_2(s) + 6 \text{NH}_3(g) \rightarrow \text{NiCl}_2 \cdot 6\text{NH}_3(s)$
 C. $\text{I}_2(g) \rightarrow 2 \text{I}(g)$
 D. $\text{ZnO}(s) + \text{CO}_2(g) \rightarrow \text{ZnCO}_3(s)$
 E. $\text{C}_2\text{H}_4(g) + \text{Cl}_2(g) \rightarrow \text{C}_2\text{H}_4\text{Cl}_2(l)$

Q11) The value of ΔH° for the reaction below is -72 kJ. Calculate amount of heat released when 80.9 grams of HBr ($M_m = 80.9 \text{ g/mol}$) is formed in this reaction. $\text{H}_2(g) + \text{Br}_2(g) \rightarrow 2\text{HBr}(g)$

- A. 144 B. 72 C. 0.44 D. 36
 E. -72

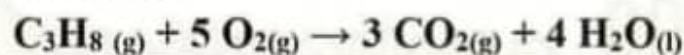
Q12) A 1.5L of an ideal gas is expanded isothermally against external pressure equal to 2.96 atm at 25°C. During expansion, the gas absorbed 250 joule of heat. What is the volume of the gas after expansion?

- A. 0.83 L B. 2.33 L C. 0.667 L D. 8.33 L
 E. 3.33 L

Q13) A correct description for the electron configuration of a vanadium atom ($_{23}\text{V}$) is

- A. $[\text{Ar}] 4s^1 3d^5$ B. $[\text{Ar}] 4s^2 3d^3$ C. $[\text{Ar}] 4s^2 3d^5$ D. $[\text{Ar}] 3d^5$
E. $[\text{Ar}] 3s^2 3d^4$

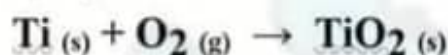
Q14) Based on the information bellow, which expression represents the enthalpy for the following reaction:



Reaction	Enthalpy of reaction
$3\text{C}(\text{s}) + 4 \text{H}_2(\text{g}) \rightarrow \text{C}_3\text{H}_8(\text{g})$	X
$\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$	Y
$2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$	Z

- A. $-X + 3Y + 2Z$ B. $X - 3Y + Z$ C. $-2X - 3Y + 2Z$ D. $-X - 3Y + 2Z$
E. $-X + 3Y - 2Z$

Q15) The combustion of titanium with oxygen produces titanium dioxide:



When 2.060 g of titanium (47.9g/mol) is combusted in a bomb calorimeter, the temperature of the calorimeter increases from 25.00 °C to 41.60 °C. If the heat capacity of the calorimeter equal 9.84 kJ/K. Calculate enthalpy of reaction for the combustion of a mole of Ti in this calorimeter.

- A. -163.3 B. -165.8 C. -326.6 D. -3798
E. -2800

Q16) Which atom has the largest first ionization energy?

- A. $_{13}\text{Al}$ B. $_{38}\text{Sr}$ C. $_{31}\text{Ga}$ D. $_{12}\text{Mg}$
E. $_{11}\text{Na}$

Q17) Which one of the following statements is true?

- A. Enthalpy is an intensive property.
B. The enthalpy change for a reaction is independent of the state of the reactants and products.
C. Enthalpy is a state function.
D. H is the value of q measured under conditions of constant volume.
E. The enthalpy change of a reaction is the reciprocal of the ΔH of the reverse reaction.

Q18) What is the mass % of carbon in dimethylsulfoxide ($\text{C}_2\text{H}_6\text{SO}$)?

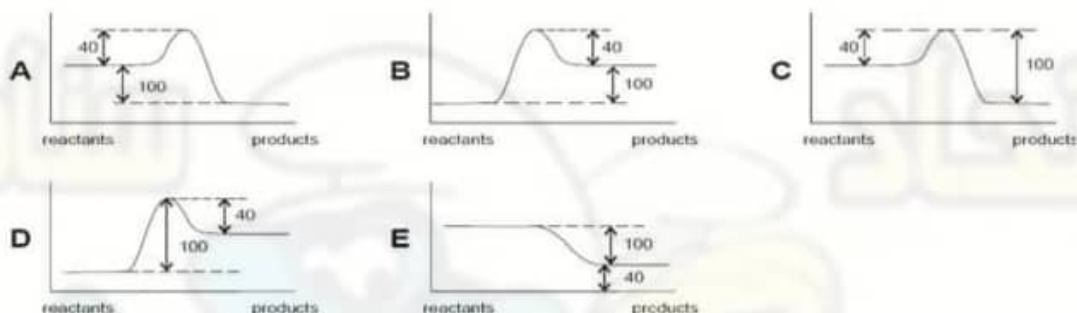
(At.wtsg/mol: C= 12, H=1, S = 32, O =16)

- A. 60.0 B. 20.5 C. 30.7 D. 7.69
E. 79.8

Q19) Under appropriate conditions, nitrogen and hydrogen undergo a combination reaction to yield ammonia: $\text{N}_2(\text{g}) + 3 \text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. The reaction percentage yield = 17.4% Calculate the actual yield of NH_3 produced if 7.1 g sample of N_2 are allowed to react with enough amount of H_2 . (At.wts : N=14, H=1 g/mol)

- A. 5.3 B. 6.7 C. 4.6 D. 1.5
E. 17.4

Q20) An endothermic reaction ($\text{A} \leftrightarrow \text{B}$) has enthalpy equal to 60 kJ, and activation energy equal to 40 kJ for its reverse reaction. Which of one of the energy diagrams shown below best describes this reaction.



Q21) A 25.00 mL of a vinegar solution is diluted to 250.0 mL. If the concentration of the diluted vinegar solution is 0.08527 M. What is the concentration of the original vinegar?

- A. 0.2578 M B. 0.5287 M C. 0.7825 M D. 0.8527 M
E. 0.9978 M

Q22) If 100 mL of water is added to 50.0 mL of 0.60 M solution of AlCl_3 . What is the final

concentration of chloride ion (Cl^-) after adding water?

- A. 1.8 M B. 0.3M C. 0.2M D. 0.6M
E. 0.9 M

Q23) What is the molar concentration of a solution prepared by dissolving 10.0 g of cobalt(II) chloride dihydrate ($\text{CoCl}_2 \cdot 2\text{H}_2\text{O}$ M.m=165.87 g/mol) in enough amount of ethanol to make a 500 mL solution.

- A. 0.121 M B. 0.263M C. 0.526 M D. 0.734 M
E. 0.972 M

Q24) For the reaction: $\text{Na}_2\text{CO}_3(\text{s}) + 2\text{HNO}_3(\text{l}) \longrightarrow 2\text{NaNO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$, how many grams of Na_2CO_3 (molar mass = 106.0 g/mol) are required to react completely with 25.0 mL of 0.155 M HNO_3 ?

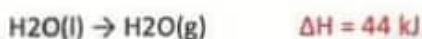
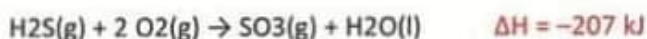
- A. 0.125 g B. 0.205 g C. 0.410 g D. 1.250 g
E. 2.050 g

Thermodynamics (worksheet)

1) Find the ΔH for the reaction below,



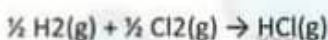
given the following reactions and subsequent ΔH values:



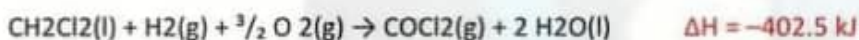
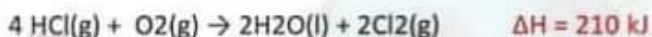
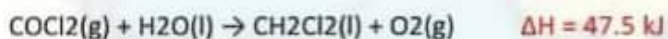
Answer: +72.5 KJ

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2) Find the ΔH for the reaction below,



given the following reactions and subsequent ΔH values:



ANSWER = -230 kJ

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3) ΔH for the reaction



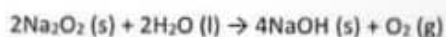
is _____ kJ, give the data below.



A) -1135 B) +35 C) +1135 D) -35 E) +355

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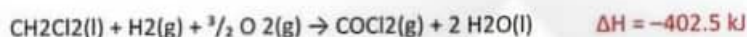
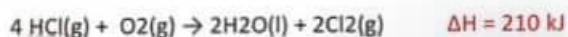
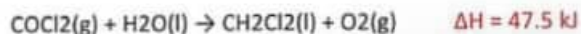
4) The value of ΔH for the reaction below is -126 kJ. The amount of heat that is released by the reaction of 25.0 g of Na_2O_2 with water is _____ kJ.



Answer: -20.19 kJ

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given the following reactions and subsequent ΔH values:



ANSWER = -230 kJ

3) ΔH for the reaction

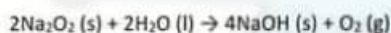


is _____ kJ, give the data below.



A) -1135 B) +35 C) +1135 D) -35 E) +355

4) The value of ΔH for the reaction below is -126 kJ. The amount of heat that is released by the reaction of 25.0 g of Na_2O_2 with water is _____ kJ.



Answer: -20.19 kJ

5) The specific heat of aluminum is $0.900 \text{ J/g}^\circ\text{C}$.

How much energy is needed to raise the temperature of a 8.50kg block of aluminum from 22.8°C to 94.6°C ? 549.3 kJ

6) A piece of metal weighing 59.047 g was heated to 100.0°C and then put it into 100.0 mL of water (initially at 23.7°C). The metal and water were allowed to come to an equilibrium temperature, determined to be 27.8°C . Assuming no heat lost to the environment, calculate the specific heat of the metal.

6. In a coffee-cup calorimeter, 100.0 g of H_2O and 100.0 mL of HCl are mixed. The HCl had an initial temperature of 44.6°C and the water was originally at 24.6°C . After mixing, the temperature of both substances is 31.3°C .

a. Was the process exothermic or endothermic? Explain.

Exothermic for HCl but endothermic for water.

b. Calculate how much heat the water lost or gained.

$$q = +2803.28 \text{ J} = 2.8 \text{ kJ}$$